Unit-1:

1. Write code demonstrating variables holding different data types: string, integer, float, Boolean.

localhost/suresh/php/unit-1/lab/DataTypesDemo.php

Integer: 10 int(10)
Float: 3.14 float(3.14)
String: Hello, PHP! string(11) "Hello, PHP!"
Boolean: 1 bool(true)
Array: Array ([0] => 1 [1] => 2 [2] => 3) array(3) { [0]=> int(1) [1]=> int(2) [2]=> int(3) }
Type of intVar: integer
Type of floatVar: double
Type of stringVar: string
Type of boolVar: boolean
Type of arrayVar: array

2. Show different ways to declare and initialize variables in PHP.

localhost/suresh/

Name: Suresh
Age: 28
Salary: 35000.5
Student: no
Skills: PHP, HTML, CSS
Institute: ABC Institute

3. Write expressions using various mathematical, comparison, and logical operators in PHP.

localhost/suresh/php/unit-1/Question-3.php

Name: Suresh Dahal
Question 3: Mathematical, Comparison, and Logical Operators

Mathematical Operators:
20 + 10 = 30
20 - 10 = 10
20 * 10 = 200
20 / 10 = 2
20 % 10 = 0

Comparison Operators:
20 == 10 : false
20 != 10 : true
20 > 10 : true
20 < 10 : false
20 >= 10 : true
20 <= 10 : false

Logical Operators:
true AND false = false
true OR false = true
NOT true = false

4. Write code demonstrating loop controls like for, while, and do-while loops in PHP.

```
C:\xampp\htdocs\suresh\php\unit-1\lab>php ControlStructuresDemo.php
a is positive
Wednesday
for: 1
for: 2
for: 3
while: 1
while: 2
while: 3
do-while: 1
do-while: 2
do-while: 3
foreach: 10
foreach: 20
foreach: 30
break and continue in for loop:
1 3
```

5. Demonstrate including external PHP files using require and include statements.

localhost/suresh/php/unit-1/lab/IncludeDemo.php

Hello, Suresh!
5
Demonstrating include vs require:
After include (script continues)
After require (this will not print if require fails)

6. WAP to demonstrate embedding PHP directly in HTML page.

localhost/suresh/php/unit-1/lab/DynamicPage.php

Welcome to Dynamic PHP Page

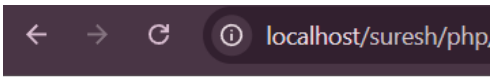
Hello, **Suresh**!

Course: *Web Programming*

Current date and time: **Thursday, February 5, 2026 04:29:54 AM**

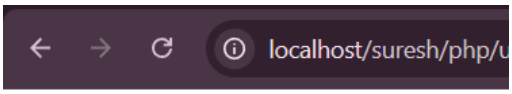
Unit-2:

1. Write a function that receives a string and checks if it's a palindrome (reads the same backward) in PHP



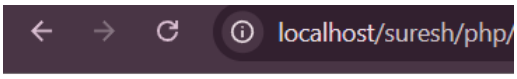
Not a Palindrome
Palindrome

2. Demonstrate how to call a function within another function in PHP.



Sum: 8

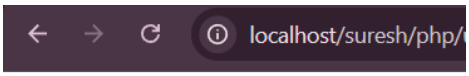
3. How can you access a global variable from within a function in PHP, Write the program.



Name: Suresh Dahal
Age: 25

Using GLOBALS array:
Name: Suresh Dahal
Age: 25

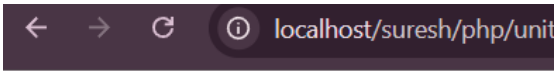
4. Write a function that swaps the values of two variables using parameter references in PHP.



Before swap:
num1 = 10
num2 = 20

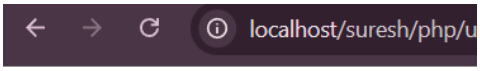
After swap:
num1 = 20
num2 = 10

5. Write a function that calculates the area of a circle and returns the result in PHP.



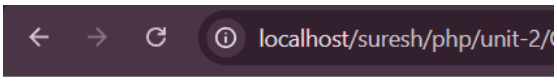
Area of circle with radius 5: 78.53975

6. Write an anonymous function that checks if a string is empty in PHP.



Is str1 empty? Yes
Is str2 empty? No

7. Generate the current date and time in different formats using PHP functions.



Current: 2026-02-04 15:17:25
Date (d/m/Y): 04/02/2026
Date (m-d-Y): 02-04-2026
Date (D, d M Y): Wed, 04 Feb 2026
Time (h:i A): 03:17 PM
Time (h:i:s): 15:17:25
Day of the week: Wednesday
Month: February
Year: 2026
Unix timestamp: 1770214645
Add 1 week: 2026-02-11
DateTime +10 days: 2026-02-14

Unit-3:

1. Demonstrate different ways to print a string in PHP.

localhost/suresh/php/u

Direct HTML: Hello, World!

Suresh Dahal
Suresh Dahal
Suresh Dahal
Suresh Dahal

Suresh Dahal

2. WAP to access individual characters in a string in PHP.

localhost/suresh/php/uni

First character: S
Eighth character: D
last 4 characters: ahal
Substring (0, 6): Suresh
Characters one by one: Suresh Dahal

3. WAP to remove whitespace from the beginning and end of a string in PHP.

localhost/suresh/php/

```
Original: '  Suresh Dahal  '

trim: 'Suresh Dahal'

ltrim: 'Suresh Dahal  '

rtrim: '  Suresh Dahal'
```

4. WAP to use the functions which are used for HTML and URL encoding in PHP.

localhost/suresh/php/unit-3/Question-4.php

htmlspecialchars: Suresh Dahal
htmlentities: Suresh Dahal
urlencode: Suresh+Dahal+%26+PHP
rawurlencode: Suresh%20Dahal%20%26%20PHP
urldecode: Suresh Dahal & PHP

Preventing XSS:
User input (unsafe): Click Me
Escaped (safe): <button onclick='alert("Hello")'>Click Me</button>

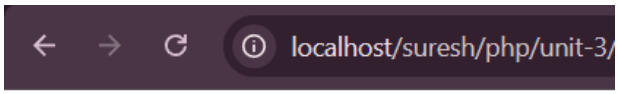
5. WAP to prevent SQL injection by escaping strings in PHP.

localhost/suresh/php/unit-3/Question-5

Original input: O'Reilly
Escaped: O\'Reilly
Escaped with addslashes: O\'Reilly

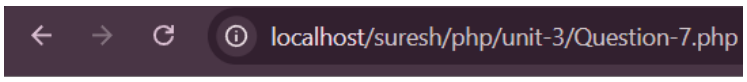
Using Prepared Statements to prevent SQL Injection
Prepared statement executed successfully.

6. Demonstrate various methods for comparing two strings in PHP.



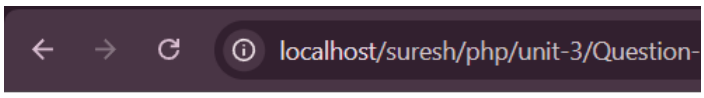
==: false
strcmp: false
strcasecmp: equal
strncmp: false
strncasecmp: equal

7. WAP to use string manipulation functions like strlen, substr, and str_replace in PHP.



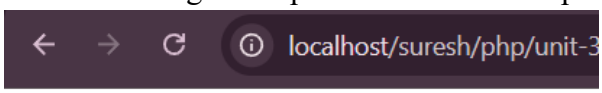
strlen: 12
substr: Suresh
str_replace: Hello Suresh Dahal
str_ireplace: Hello BIT
uppercase: SURESH DAHAL LOVES PHP. LOVES CODING.
lowercase: suresh dahal loves php. loves coding.
position of 'loves': 13
position of 'Loves' (case-insensitive): 13
word count: 6

8. WAP to use strpos and stristr functions for searching within strings in PHP.



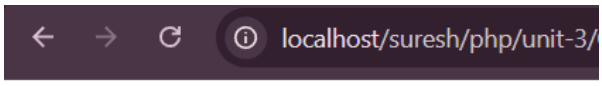
String: Suresh Dahal is learning PHP.
Position of 'Suresh': 0
Position of 'php' (case-insensitive): 25
Substring before 'learning': Suresh Dahal is
Substring from 'is': is learning PHP.

9. WAP to use regular expressions in PHP for pattern matching in strings.



Found email: suresh@example.com
Email not found.

10. WAP to create an array and access its elements in PHP.



Found 'cherry' at index: 2
First element: red
Last element: blue
Name: Suresh Dahal

11. WAP to add elements to an indexed array dynamically in PHP.

localhost/suresh/php/unit-3

Apple
Ball
Cat
Item 1
Item 2
Item 3

12. WAP to create and access elements in an associative array in PHP.

localhost/suresh/php/unit-

Name: Suresh Dahal
Age: 25

13. WAP to use an array to store user details (name, age, email) in PHP.

localhost/suresh/php/unit-3/Question-1

User Details:
Name: Suresh Dahal
Age: 30
Email: suresh@example.com

All Users:
Suresh Dahal, 25, suresh@example.com
Asha Rai, 21, asha@example.com

14. WAP to create and manipulate a two-dimensional array in PHP.

localhost/suresh/php/unit-3/Que

Original Matrix:
1 2 3
4 5 6
7 8 9

Matrix after manipulation:
1 2 3
4 5 99
7 8 9

15. WAP to extract multiple values from an array in PHP.

localhost/suresh/php/unit-3/Question-15.php

Method 1: list()
a = 1, b = 2, c = 3

Method 2: Array destructuring
x = 1, y = 2, z = 3

Method 3: array_slice()
Array ([0] => 2 [1] => 3 [2] => 4)
Method 4: array_map() for specific keys
Array ([0] => 1 [1] => 3 [2] => 5)
Method 5: foreach for specific indices
Array ([0] => 2 [1] => 4)

16. WAP to convert a variable to an array and vice-versa in PHP.

localhost/suresh/php/unit-3/Question-16.php

Variable to array methods:

Using type cast:
Array ([0] => Hello World)

Using explode:
Array ([0] => Hello [1] => World)

Using str_split:
Array ([0] => 1 [1] => 2 [2] => 3)

Array to variables methods:

Using extract():
first = Suresh, last = Dahal

Using implode():
Hello World

Using list():
fruit1 = apple, fruit2 = banana, fruit3 = cherry

17. WAP to use loops to traverse both indexed and associative arrays in PHP.

localhost/suresh/php/unit-3/Question-17.php

Traversing indexed array with for loop:
Element 0: Apple
Element 1: Banana
Element 2: Cherry

Traversing indexed array with foreach loop:
Value: Apple
Value: Banana
Value: Cherry

Traversing associative array with foreach loop:
name: Suresh
age: 25
city: Jhapa

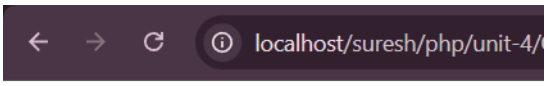
Traversing associative array with for loop:
name: Suresh
age: 25
city: Jhapa

Unit-4:

1. Write PHP code to create an object of a class named "Book" with properties like title, author, and publication year.

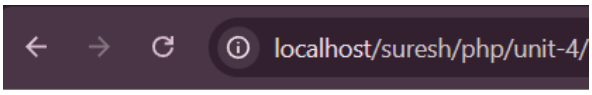
The Unseen Heartbeat by Suresh Dahal (2028)

2. Demonstrate how to access the properties of an object in PHP. Include examples of both public and private properties.



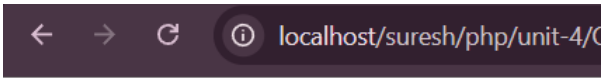
Public: public value
Private via getter: private value
Private via reflection: private value

3. Write a PHP class declaration for a "Person" class with attributes like name, age, and address.



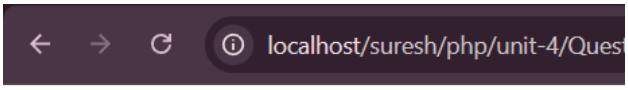
Name: Suresh Dahal
Age: 25
Address: Kathmandu, Nepal

4. Write a PHP program which uses a class with a constructor.



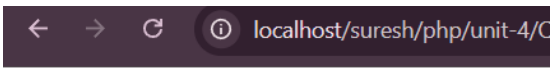
Toyota

5. Write a PHP program which uses an anonymous class.



Hello from anonymous

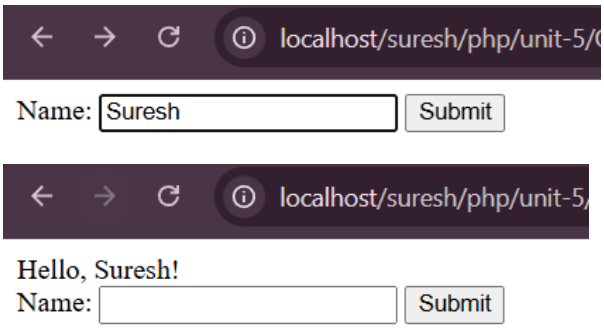
6. How can reflection be used in PHP to examine and manipulate classes and objects? WAP to demonstrate the given concept.



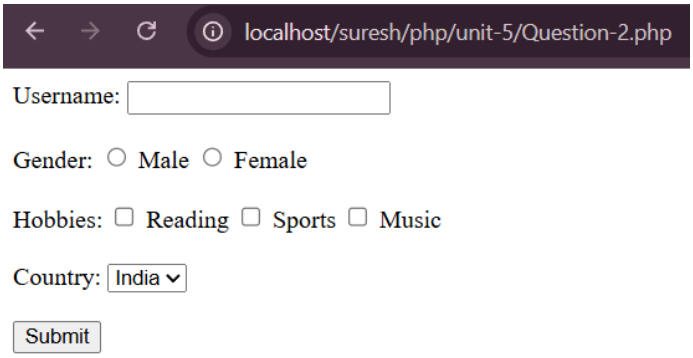
Class: Sample
x
x before: 10
x after: 11

Unit-5:

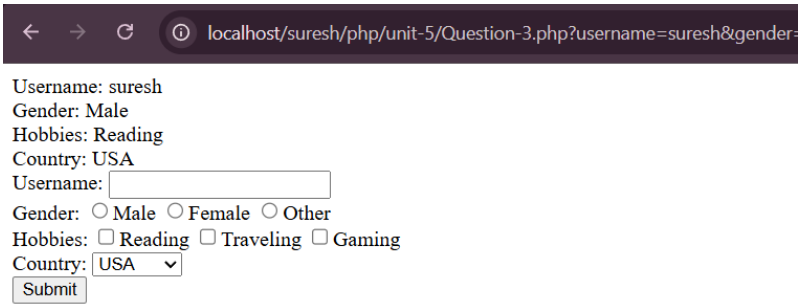
1. WAP to perform “action" and "method" attributes in an HTML form. How are they utilized in PHP form processing?



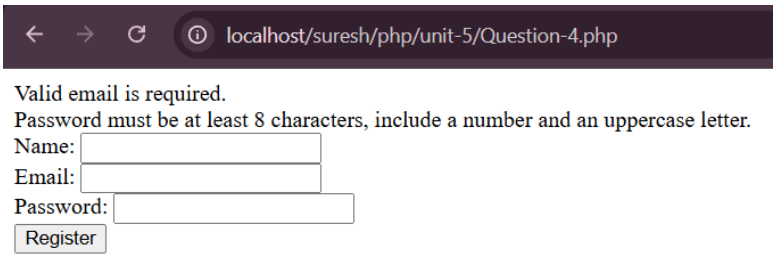
2. Implement a form in PHP that includes various input types such as text, radio buttons, checkboxes, and a select dropdown.



3. Demonstrate how to access individual form field values submitted through a POST/GET request in PHP.



4. Write PHP code to validate form input for a registration form, checking for empty fields, valid email addresses, and secure passwords.



5. Write PHP code to set a custom response header, such as "Content-Type" or "Location," in order to control the behavior of the client browser.

localhost/suresh/php/unit-5/Question-5.php

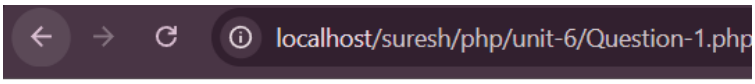
Username: suresh Submit

localhost/suresh/php/unit-5/Question-5.php


```
{
    "message": "Welcome, suresh!"
}
```

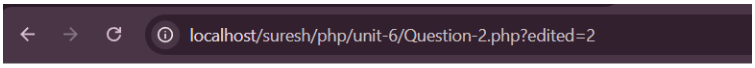
Unit-6:

1. WAP to provide the steps involved in connecting to a MySQL database using PHP. Provide a code using mysqli and PDO extensions.



Connected successfully using MySQLi.
Connected successfully using PDO.

2. Create an HTML form for a simple "Product" database table, including fields for product name, description, price, and quantity. Implement CRUD operations.



Add Product:

Name: Mouse

Price: 130

Quantity: 30

Description: 3 button wired mouse

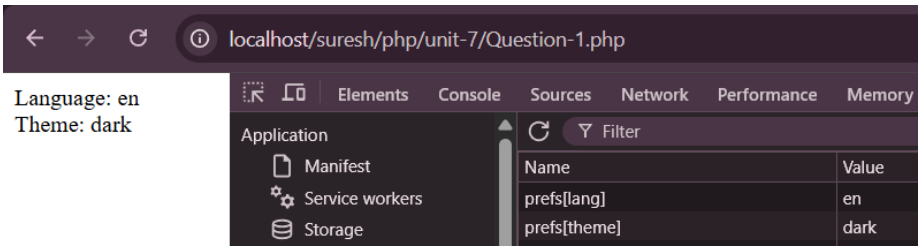
Update Product

Product List:

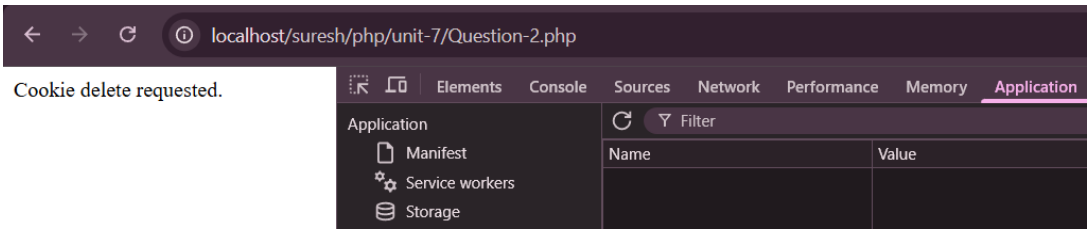
ID	Name	Price	Quantity	Description	Actions
2	Mouse	130	30	3 button wired mouse	Edit | Delete

Unit-7:

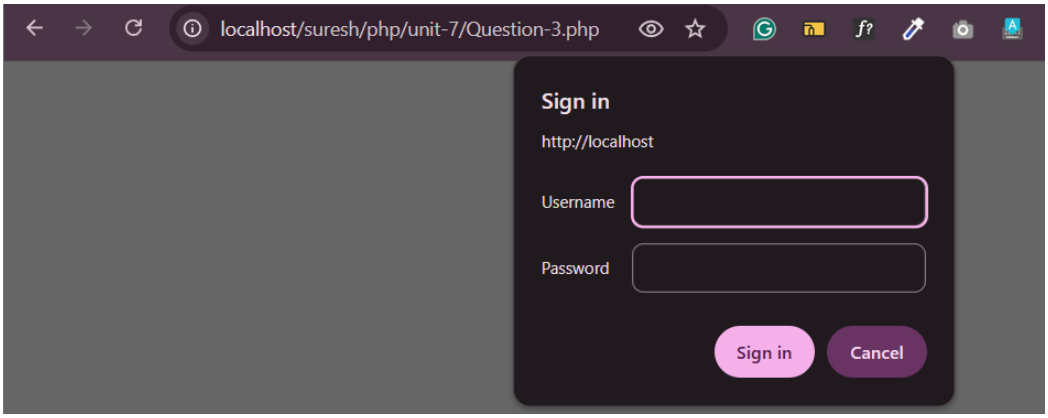
1. Create a PHP script that sets a cookie with a user's preferences, such as language or theme. Retrieve and use this cookie in subsequent visits.

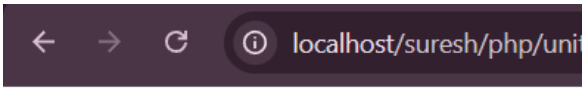


2. Write PHP code to delete a cookie on the client-side and explain the implications of doing so.



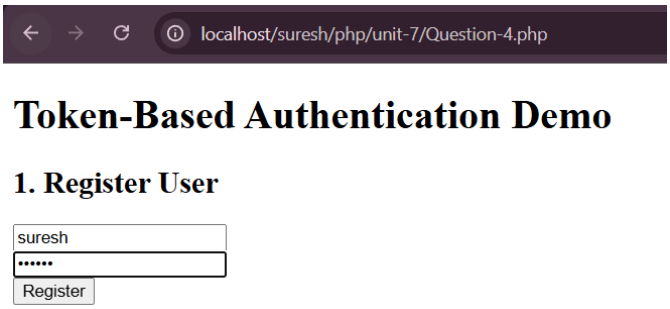
3. Implement HTTP Basic Authentication in PHP for a protected area of a website.





Hello, admin

4. How can you secure an API using token-based authentication in PHP? Provide a step-by-step explanation with PHP code.



localhost/suresh/php/unit-7/Question-4.php

Token-Based Authentication Demo

User 'suresh' registered successfully!

1. Register User

Username
Password
Register

localhost/suresh/php/unit-7/Question-4.php

Token-Based Authentication Demo

Login successful! Token generated (expires in 1 hour)
Generated Token:
29c24f87f7033287715111d6869b779236ff22700f917c745733665291308ca6

1. Register User

Username
Password
Register

2. Login (Get Token)

Username
Password
Login

3. Access Protected Resource

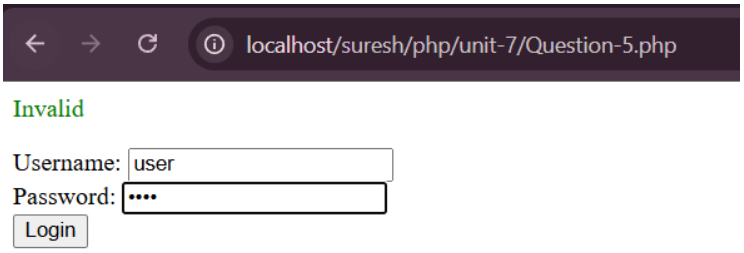
29c24f87f7033287715111d6869b779236ff22700f917c745733665291308ca6
Access Protected API

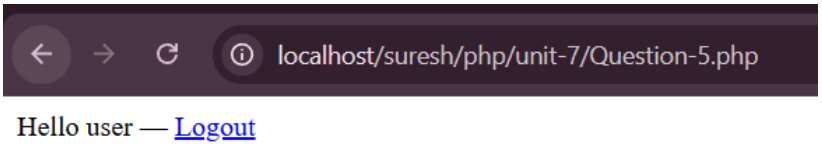
localhost/suresh/php/unit-7/Question-4.php

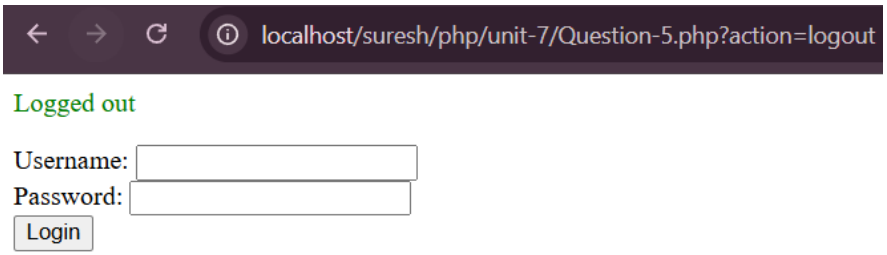
Token-Based Authentication Demo

Access granted! Welcome suresh

5. Create a simple login system using PHP sessions. Include functionality for login, logout, and checking if a user is authenticated.







Unit-8:

1. WAP to use php.ini file in PHP configuration program.

localhost/suresh/php/unit-8/Question-1.php

Loaded php.ini: C:\xampp\php\php.ini
display_errors: 1

2. WAP to use E_ERROR and E_WARNING error types in PHP.

localhost/suresh/php/unit-8/Question-2.php

E_WARNING: include(missingfile.php): Failed to open stream: No such file or directory (Line: 11)
E_WARNING: include(): Failed opening 'missingfile.php' for inclusion (include_path='C:\xampp\php\PEAR') (Line: 11)
1. Division by zero:
Fatal error: Uncaught DivisionByZeroError: Division by zero in C:\xampp\htdocs\suresh\php\unit-8\Question-2.php:15 Stack trace: #0 {main} thrown in **C:\xampp\htdocs\suresh\php\unit-8\Question-2.php** on line **15**

3. WAP to use a try-catch block to handle exceptions in PHP.

localhost/suresh/php/unit-8/Question-3.php

Caught: This is an E_ERROR simulation.
End of Script.

4. WAP to use @ symbol before a function call in PHP.

localhost/suresh/php/unit-8/Question-4.php

Continue even if unlink failed

5. WAP to trigger a custom error condition using the trigger_error() function.

localhost/suresh/php/unit-8/Question-5.php

Custom handler [1024]: A custom notice

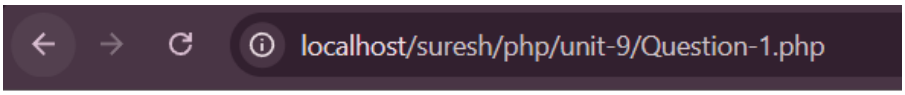
6. WAP to configure PHP to log errors to a file.

localhost/suresh/php/unit-8/Question-6.php

Notice: Test error for log file in **C:\xampp\htdocs\suresh\php\unit-8\Question-6.php** on line **6**
Errors logged to: C:\xampp\htdocs\suresh\php\unit-8/php_errors.log

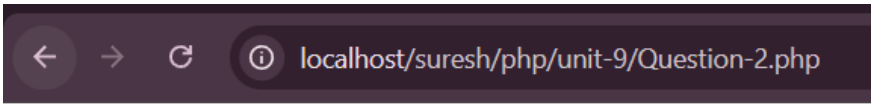
Unit-9:

1. Demonstrate how to open a file for writing in PHP using fopen() and write a simple string to it.



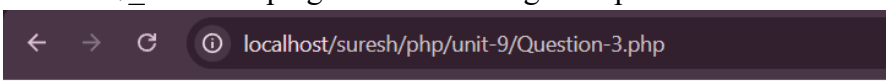
Wrote to C:\xampp\htdocs\suresh\php\unit-9/sample.txt

2. WAP to read the entire contents of a text file into a string variable in PHP.



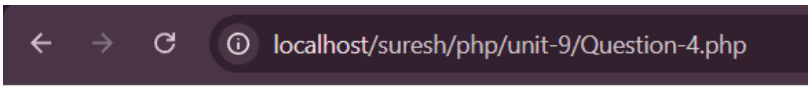
Hello from PHP

3. WAP to use $_FILES superglobal in handling file uploads in PHP.



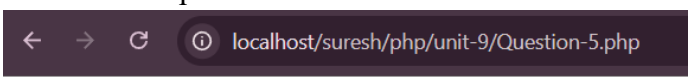
Uploaded to C:\xampp\htdocs\suresh\php\unit-9/uploads/bba-nepal.vercel.app_.png
Choose file No file chosen Upload

4. WAP to read and parse a CSV file in PHP.



sn name roll
1 Suresh 23
2 Anita 21
3 John 22

5. WAP to read and parse a JSON file in PHP.



Name: Suresh Dahal
Roll: 23

Name: Ram Dahal
Roll: 45

Name: Shyam Dahal
Roll: 67

LAB-2: Create an HTML form for a simple "Product" database table, including fields for product name, description, price, and quantity. Implement CRUD operations.

```
<?php
    $db = new mysqli('localhost', 'root', '', 'inventory');
    if($db->connect_error)
      die("Connection failed: " . $db->connect_error);
    if(isset($_POST['add'])){
      $name = $_POST['name'] ?? '';
      $price = floatval($_POST['price'] ?? 0);
      $quantity = intval($_POST['quantity'] ?? 0);
      $description = $_POST['description'] ?? '';
      $stmt = $db->prepare("INSERT INTO products (name, price, quantity, description)
    VALUES (?, ?, ?, ?)");

      if($stmt){
        $stmt->bind_param("sdis", $name, $price, $quantity, $description);
        if($stmt->execute()){
          echo "Product added successfully.";
        } else
          echo "Error: " . $stmt->error;
        $stmt->close();
      } else {
        echo "Error preparing statement: " . $db->error;
      }

      header("Location: $_SERVER[PHP_SELF]");
      exit();
    }

    if(isset($_GET['deleted'])){
```

```
    $id = intval($_GET['deleted']);
    $stmt = $db->prepare("DELETE FROM products WHERE id = ?");
    if($stmt){
        $stmt->bind_param("i", $id);
        if($stmt->execute()){
            echo "Product deleted successfully.";
        } else {
            echo "Error: " . $stmt->error;
        }
        $stmt->close();
    } else {
        echo "Error preparing statement: " . $db->error;
    }
    header("Location: $_SERVER[PHP_SELF]");
    exit();
}

if(isset($_POST['update'])){
    $id = intval($_POST['id'] ?? 0);
    $name = $_POST['name'] ?? '';
    $price = floatval($_POST['price'] ?? 0);
    $quantity = intval($_POST['quantity'] ?? 0);
    $description = $_POST['description'] ?? '';

    $stmt = $db->prepare("UPDATE products SET name = ?, price = ?, quantity = ?,
description = ? WHERE id = ?");
    if($stmt){
        $stmt->bind_param("sdisi", $name, $price, $quantity, $description, $id);
        if($stmt->execute()){
```

```
                echo "Product updated successfully.";
            } else {
                echo "Error: " . $stmt->error;
            }
            $stmt->close();
        } else {
            echo "Error preparing statement: " . $db->error;
        }
        header("Location: $_SERVER[PHP_SELF]");
        exit();
    }
    $edit = null;
    if(isset($_GET['edited'])){
        $eid = intval($_GET['edited']);
        $stmt = $db->prepare("SELECT * FROM products WHERE id = ?");
        if($stmt){
            $stmt->bind_param("i", $eid);
            $stmt->execute();
            $res2 = $stmt->get_result();
            $edit = $res2->fetch_assoc();
            $stmt->close();
        }
    }
?>
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Product</title>
</head>
<body>
    <h1>Add Product: </h1>
    <form action="" method="post">
        Name: <input type="text" value="<?php echo htmlspecialchars($edit['name'] ??
"); ?>" name="name" required><br><br>
        Price: <input type="number" step="0.01" value="<?php echo
htmlspecialchars($edit['price'] ?? "); ?>" name="price" required><br><br>
        Quantity: <input type="number" value="<?php echo
htmlspecialchars($edit['quantity'] ?? "); ?>" name="quantity" required><br><br>
        Description: <textarea name="description" required><?php echo
htmlspecialchars($edit['description'] ?? "); ?></textarea><br><br>
        <?php if($edit): ?>
            <input type="hidden" name="id" value="<?php echo $edit['id']; ?>">
            <input type="submit" name="update" value="Update Product">
        <?php else: ?>
            <input type="submit" name="add" value="Add Product">
        <?php endif; ?>
    </form>


    <?php
        $res = $db->query("SELECT * FROM products");
        if($res->num_rows > 0):
    ?>
        <h2>Product List: </h2>
        <table border="1" cellpadding="10">
            <tr>
                <th>ID</th>
                <th>Name</th>
```

```
                <th>Price</th>
                <th>Quantity</th>
                <th>Description</th>
                <th>Actions</th>
            </tr>
            <?php while($row = $res->fetch_assoc()): ?>
                <tr>
                    <td><?php echo $row['id']; ?></td>
                    <td><?php echo $row['name']; ?></td>
                    <td><?php echo $row['price']; ?></td>
                    <td><?php echo $row['quantity']; ?></td>
                    <td><?php echo $row['description']; ?></td>
                    <td>
                        <a href="?edited=<?php echo $row['id']; ?>">Edit</a> |
                        <a href="?deleted=<?php echo $row['id']; ?>" onclick="return
confirm('Are you sure?')">Delete</a>
                    </td>
                </tr>
            <?php endwhile; ?>
        </table>
    <?php endif; ?>
</body>
</html>
```

LAB-4: How can you secure an API using token-based authentication in PHP? Provide a step-by-step explanation with PHP code.

```
<?php
    session_start();
    $db = new mysqli('localhost', 'root', '', 'college');
    $db->query("CREATE TABLE IF NOT EXISTS users (
      id INT AUTO_INCREMENT PRIMARY KEY,
      username VARCHAR(100) UNIQUE NOT NULL,
      password VARCHAR(255) NOT NULL
    )");

    $db->query("CREATE TABLE IF NOT EXISTS api_tokens (
      id INT AUTO_INCREMENT PRIMARY KEY,
      user_id INT NOT NULL,
      token_hash VARCHAR(64) NOT NULL,
      expires_at DATETIME NOT NULL,
      FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
    )");

    $isApi = isset($_GET['action']) && $_GET['action'] !== 'view';
    if($isApi) {
      header('Content-Type: application/json');
    }

    function generateToken() {
      return bin2hex(random_bytes(32));
    }

    function storeToken($db, $userId, $token, $ttl = 3600) {
```

```
    $hash = hash('sha256', $token);
    $stmt = $db->prepare("INSERT INTO api_tokens (user_id, token_hash, expires_at)
VALUES (?, ?, DATE_ADD(NOW(), INTERVAL ? SECOND))");
    $stmt->bind_param("isi", $userId, $hash, $ttl);
    $stmt->execute();
    $stmt->close();
}

function authenticate($db, $token = null) {
    if(!$token) {
        $headers = getallheaders();
        $auth = $headers['Authorization'] ?? null;
        if(!$auth || stripos($auth, 'Bearer ') !== 0) return null;
        $token = substr($auth, 7);
    }

    $hash = hash('sha256', $token);
    $stmt = $db->prepare("SELECT user_id FROM api_tokens WHERE token_hash =
? AND expires_at > NOW()");
    $stmt->bind_param("s", $hash);
    $stmt->execute();
    $result = $stmt->get_result();
    $row = $result->fetch_assoc();
    $stmt->close();

    return $row ? $row['user_id'] : null;
}

$action = $_GET['action'] ?? '';
```

```
$message = $_SESSION['message'] ?? '';
$token = $_SESSION['token'] ?? '';
unset($_SESSION['message'], $_SESSION['token']);

switch($action) {
    case 'register':
        $username = $_POST['username'] ?? '';
        $password = $_POST['password'] ?? '';
        $hashedPassword = password_hash($password, PASSWORD_BCRYPT);

        $stmt = $db->prepare("INSERT INTO users (username, password) VALUES (?,
?)");
        $stmt->bind_param("ss", $username, $hashedPassword);

        if($stmt->execute()) {
            $_SESSION['message'] = "User '$username' registered successfully!";
        } else {
            $_SESSION['message'] = "Username already exists";
        }
        $stmt->close();
        header("Location: $_SERVER[PHP_SELF]");
        exit();

    case 'login':
        $username = $_POST['username'] ?? '';
        $password = $_POST['password'] ?? '';

        $stmt = $db->prepare("SELECT id, password FROM users WHERE username =
?");
        $stmt->bind_param("s", $username);
```

```
        $stmt->execute();
        $user = $stmt->get_result()->fetch_assoc();
        $stmt->close();

        if(!$user || !password_verify($password, $user['password'])) {
            $_SESSION['message'] = "Invalid credentials";
        } else {
            $token = generateToken();
            storeToken($db, $user['id'], $token, 3600);
            $_SESSION['message'] = "Login successful! Token generated (expires in 1
hour)";
            $_SESSION['token'] = $token;
        }
        header("Location: $_SERVER[PHP_SELF]");
        exit();

    case 'protected':
        $inputToken = trim($_POST['token'] ?? '');

        if(empty($inputToken)) {
            $_SESSION['message'] = "Please provide a token";
        } else {
            $userId = authenticate($db, $inputToken);
            if(!$userId) {
                $_SESSION['message'] = "Invalid or expired token. Please login again.";
            } else {
                $stmt = $db->prepare("SELECT username FROM users WHERE id = ?");
                $stmt->bind_param("i", $userId);
                $stmt->execute();
```

```
                $user = $stmt->get_result()->fetch_assoc();
                $stmt->close();
                $_SESSION['message'] = "Access granted! Welcome " . $user['username'];
            }
        }
        header("Location: $_SERVER[PHP_SELF]");
        exit();
    }
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Token Authentication Demo</title>
    </head>
    <body>
        <h1>Token-Based Authentication Demo</h1>

        <?php if($message): ?>
            <div class="message"><?php echo $message; ?></div>
        <?php endif; ?>

        <?php if($token): ?>
            <div class="token"><strong>Generated Token:</strong><br><?php echo
    $token; ?></div>
        <?php endif; ?>

        <div class="section">
            <h2>1. Register User</h2>
            <form method="POST" action="?action=register">
```

```
            <input type="text" name="username" placeholder="Username" required><br>
            <input type="password" name="password" placeholder="Password"
        required><br>
            <button type="submit">Register</button>
        </form>
    </div>

    <div class="section">
        <h2>2. Login (Get Token)</h2>
        <form method="POST" action="?action=login">
            <input type="text" name="username" placeholder="Username" required><br>
            <input type="password" name="password" placeholder="Password"
        required><br>
            <button type="submit">Login</button>
        </form>
    </div>

    <div class="section">
        <h2>3. Access Protected Resource</h2>
        <form method="POST" action="?action=protected">
            <input type="text" name="token" placeholder="Paste token here" required
        style="width: 90%;"><br>
            <button type="submit">Access Protected API</button>
        </form>
    </div>
    </body>
</html>
<?php $db->close(); ?>
```